

Jadhav Rohit

NIT Silchar (ECE), Final Year

Backend Engineer with AI Infrastructure experience

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SUMMARY

Backend Software Engineer with hands-on experience building scalable cloud-native applications, AI-powered backend systems, and production-ready deployment pipelines. Experienced in FastAPI, Node.js, Docker, Kubernetes, Redis, MongoDB, and AWS, with practical exposure to LLM deployment, edge AI inference, and distributed system fundamentals. Passionate about Enterprise AI, Retrieval-Augmented Generation (RAG), and building intelligent developer tools that combine modern software engineering with Generative AI.

EDUCATION

National Institute of Technology, Silchar <i>B.Tech in Electronics and Communication Engineering CGPA: 7.77</i>	Assam, India 2023 – 2027
Jawahar Navodaya Vidyalaya, Bharuch <i>Class XII – 87.4% Class X – 81.8%</i>	Gujarat, India 2020 – 2022
CSRL Super 30 <i>Competitive IIT-JEE Residential Program – Selected from across India</i>	Delhi, India 2022 – 2023

INTERNSHIP

Research Intern – S.N. Bose Research Centre, NIT Silchar <i>Edge AI YOLOv5 Raspberry Pi 4 TFLite OpenCV Python</i>	NIT Silchar June 2026 – July 2026
- Deployed real-time road anomaly detection on Raspberry Pi 4 using quantized YOLOv5 Nano, achieving 88.2% accuracy and 14–16 FPS on edge hardware. - Converted model to INT8 TFLite format via Google AI Edge LiteRT, reducing RAM by 75% and doubling inference speed over FP32 baseline. - Engineered 5-stage real-time video pipeline sustaining continuous operation below 50°C CPU with 30°C+ thermal headroom on the Pi. - Patched a Python 3.13 runtime crash via dynamic imp module injection and implemented temporal cooldown logic to eliminate CSV I/O bottlenecks.	

PROJECTS

(1) SocietyPay – AI-Powered Society Management Platform <i>React Native Node.js MongoDB Redis Docker AWS EC2 FastAPI Scikit-learn</i>	GitHub ↗ 2025 – 2026
- Built production full-stack mobile platform with React Native, Node.js, MongoDB Atlas, and Python ML microservice via FastAPI. - Implemented JWT authentication, RBAC, bcrypt password hashing, and API rate limiting to secure all endpoints against unauthorized access. - Integrated Redis caching to reduce MongoDB query load and BullMQ for asynchronous background job processing of notifications and payment workflows. - Containerized with Docker, deployed on AWS EC2 with Nginx reverse proxy, and configured GitHub Actions CI/CD pipeline. - Built NLP complaint classifier (Naive Bayes + CountVectorizer) and Isolation Forest visitor anomaly detection served as REST microservices. - Integrated Razorpay payment gateway for secure maintenance fee collection with auto-verification and real-time secretary dashboard.	
(2) Open Source LLM on Kubernetes – End-to-End AI Deployment <i>Kubernetes Docker FastAPI Ollama Mistral 7B Prometheus Grafana</i>	GitHub ↗ 2026
- Deployed Mistral 7B via Ollama on Kubernetes with FastAPI backend and React chatbot frontend. - Configured Prometheus + Grafana monitoring; debugged pod networking and ImagePullBackOff errors end-to-end.	

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, Python, SQL, HTML, CSS Frontend & Mobile: React Native, React, EJS Backend: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, Redis, BullMQ, Rate Limiting, Nginx Databases: MongoDB, MongoDB Atlas, Mongoose, Redis Cloud, PostgreSQL AI / ML: Scikit-learn, Naive Bayes, Isolation Forest, NLP, YOLOv5, TFLite, OpenCV, Pandas, NumPy, Ollama, Mistral 7B DevOps & Cloud: Docker, Kubernetes, AWS EC2, GitHub Actions CI/CD, Prometheus, Grafana, Helm Tools & DSA: Git, GitHub, Linux, Postman, VS Code 200+ LeetCode Problems (Arrays, Trees, Graphs, Hashing, DP)

POSITIONS OF RESPONSIBILITY

Illuminits Society, NIT Silchar <i>Technical Head – Managed tech infrastructure, mentored juniors in full-stack development</i>	NIT Silchar 2024 – Present
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ACHIEVEMENTS

- Selected for CSRL Super 30 – national-level IIT-JEE residential program Published blog on Kubernetes + LLM deployment at llm-deployment.hashnode.dev - SOF Bronze Medalist – District Level Company Sergeant Major (CSM), NCC NIT Silchar	
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